Evaluation and Counseling of Candidates

11

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11.1 Evaluation of Candidates and Risk Factors for HCT

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11.1.1 Introduction

The evaluation of candidates and the analysis of individual risk factors for hematopoietic cell transplantation (HCT) allow establishing four fundamental aspects of the procedure:

1. The indication for HCT
2. Providing proper information to the patient
3. Choosing the best donor, conditioning, and post-HCT immunosuppression
4. Conducting benchmarking and comparative studies between centers

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11.1.2 Candidates' Evaluation Workflow

11.1.2.1 The First Visit

The most relevant aspects to include in the first visit are as follows:

- Evaluating medical history (both past and present) and conducting a physical examination (see Sect. 11.1.2.4)
- Reviewing diagnostic tests (in referred patients)
- Revaluating human leukocyte antigen (HLA) typing of the patient and potential donors (in the case of allo-HCT)
- Providing preliminary information on:
  - Therapeutic options and results
  - The HCT procedure
  - Possible complications and side effects (see specific chapters in Part V)
- Scheduling reevaluation of the current status of the disease (see Sect. 11.1.3)
- Scheduling visits with a radiation therapist (in the case of total body irradiation (TBI)), a dentist, a gynecologist, a blood bank (for a list of blood/platelet donors), an HCT unit supervisor nurse, etc.
- Obtaining the patient's signature on the informed consent form for HCT and for procurement of hematopoietic stem cells (HSCs) (in the case of auto-HCT)

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A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_11

11.1.2.2 Pre-Apheresis Visit

The important aspects involved in the pre-apheresis visit include the following:

- Assessing the results of pretransplant tests
- Providing complete information on the procedure
- In the case of peripheral blood stem cells (PBSCs), assessing the status of venous accesses. Programming central venous catheter (CVC) (if necessary) and the mobilization schedule
- Scheduling a preanesthetic visit in the case of bone marrow (BM)
- Program manipulation of HCT (if applicable) and/or cryopreservation

11.1.2.3 The Last Visit Before Admission

This involves the following aspects:

- Obtaining the final and complete patient information (see Sect. 11.1.2.5)
- Reviewing pretransplant tests (see Sect. 11.1.3)
- Scheduling admission and conditioning therapy
- If necessary, then scheduling CVC placement
- In the case of allo-HCT, confirming that the donor's evaluation is correct and that there are no contraindications for donation (see Chap. 12)
- In the case of auto-HCT, confirming that the cryopreserved cellularity is correct
- Submitting donor and recipient information to the blood bank (group, cytomegalovirus (CMV) serology, previous transfusions, etc.)
- In the case of TBI, confirming that the dosimetry has been carried out and that the radiotherapy (RT) has been programmed
- Confirming storage of patient and donor samples for serotheque and cellular library

11.1.2.4 Medical History

The following patient information should be collected:

Medical background; childhood illnesses and vaccines; allergies and adverse drug reactions; surgical interventions (previous anesthesia); medications not related to the basic process; previous transfusion history, family tree, and relevant family history; and, in women, menarche/menopause, pregnancy and childbirth, contraceptive methods used, last menstrual period, and gynecological checkups

Travel to malaria, trypanosomiasis, and human T-lymphotropic virus type I/II (HTLV-I/II) endemic areas

Previous relevant infections, including coronavirus disease 2019 (COVID-19) and vaccination status

All data about the current illness, such as:

- The date of disease onset and initial symptomatology
- The methodology used (staging)
- Radiotherapy treatments (doses and dates) conducted
- Other treatments undertaken
- Disease status

Social aspects, such as:

- Alcoholism and other drug use
- Habits
- Place of accommodation (whether close to the center) and means of transport
- Role of family members
- Cultural, economic, and intellectual aspects

11.1.2.5 Information to Provide (See Detailed Information in the Counseling Section)

Ask the patient (privately) if he or she wishes someone else to be present for the session. For adolescents, follow the rules of each country, thus respecting the right of information. Transmit as much information as possible in writing. She/he must be informed about:

- The most frequent early and late complications (see specific chapters in Parts V and VI), including graft failure, gastrointestinal (GI) complications, alopecia, sinusoidal obstruction syndrome/ veno-occlusive disease (SOS/VOD), acute graft-versus-host disease

(GvHD), early infections, chronic GvHD, late infections, relapse of the disease, infertility, endocrine complications, neoplasms, and other adeverse events
- Serious potential complications (intensive care unit (ICU) admissions) and possibility of death, the availability of an advanced healthcare directive registry, and the need to appoint a person to make decisions in case the patient is unable to do so.
- Estimated duration of admission
- Most frequent complications on discharge, outpatient follow-up, likelihood of readmission, and need for caregivers at discharge

11.1.3 Pretransplant Evaluation

All the following studies must be performed within 30 days prior to the HCT, except the assessment of the baseline disease status (7–15 days) and the pregnancy test (7 days):

- Complete blood count (CBC) and basic coagulation; complete biochemistry (including ferritin); blood type and Rh/irregular antibodies; dosage of immunoglobulins (Igs); serology CMV, Epstein–Barr virus (EBV), Herpes simplex virus (HSV), varicella zoster virus (VVZ), toxoplasma, syphilis, hepatitis B surface antigen (HBsAg), hepatitis B core antibody (HBcAb), and HBsAb (Hepatitis B surface antibody) (HTLV-I/II and Chagas disease according to the patient's origin); nucleic acid test (NAT) for hepatitis B virus (HCV), hepatitis B virus (HBV), and human immunodeficiency virus (HIV); pregnancy test
- Chest X-ray; respiratory function tests (including forced expiratory volume in 1 s (FEV1) and diffusing capacity of the lung for carbon monoxide (DLCO)); electrocardiogram; echocardiogram or isotopic ventriculography (depending on the previous treatment)
- Reevaluation of the minimal residual disease (MRD) (see specific chapters in Part IX).
- Dental evaluation; gynecological evaluation; psychological/psychiatric evaluation
- Nutritional assessment
- HLA typing (recheck) (see Chap. 9).

11.1.4 Risk Assessment

11.1.4.1 Individual Risk Factors

There are a group of variables that have a prognostic value in all predictive models.

Variables	High risk
Age	Older. Do not use as a single criterion. Relative importance
General condition	Karnofsky index <80%
Disease	Not in remission. See specific chapters
Type of donor	Others rather than HLA-identical siblings
HLA compatibility	Any HLA-A, HLA-B, HLA-C, and DRB1 difference
CMV serology	Different serology than the donor
Donor	Age > 35–40 years For a male recipient, a female donor (especially if multiparous)
Interval diagnosis of HCT	Prolonged (relevant in chronic myeloid leukemia (CML) and Severe aplastic anemia (SAA))
Comorbidities	See HCT-CI model
Iron overload	Present
Experience of the center	Non-JACIE/FACT-accredited centers

HCT-CI HCT-specific Comorbidity Index, *JACIE* Joint Accreditation Committee of International Society for Cell and Gene Therapy-European Society for Blood and Marrow Transplantation (ISCT-European Society for Blood and Marrow Transplantation (EBMT)), *FACT* Foundation for the Accreditation of Cellular Therapy

11.1.4.2 Predictive Models

The Disease Risk Index (DRI)

The Disease Risk Index categorizes people according to disease type and its status at the time of HCT. It does not take into account factors such as age or comorbidities. This score index classifies the disease into four prognostic groups and anticipates the overall survival, progression-free survival, cumulative incidence of relapse, and cumulative incidence of non-relapse mortality (see Table 11.1) (Armand et al. 2012, 2014).

Table 11.1 The disease risk index (Armand et al. 2012, 2014)

Risk	Disease
Low	AML with favorable cyt., CLL, CML, indolent B-cell NHL
Intermediate	AML intermediate cyt., MDS intermediate cyt., myeloproliferative neoplasms, MM, HL, DLBCL/ transformed indolent B-NHL, MCL, T-cell lymphoma nodal
High	AML adverse cyt., MDS adverse cyt., T-cell lymphoma extranodal

Risk	Stage
Low	CR1, CR ≥ 2, PR1, untreated, CML CP, PR ≥ 2 (if RIC)
High	PR ≥ 2 (if MAC), induction failure, active relapse, CML AP or BP

Disease risk	Stage risk	Overall risk	OS at 4 years
Low	Low	Low	64% (56–70%)
Low	High	Intermediate	46% (42–50%)
Intermediate	Low		
Intermediate	High	High	26% (21–31%)
High	Low		
High	High	Very high	6 (0–21%)

Adapted from Armand et al. (2012)
AML acute myeloid leukemia, *MM* multiple leukemia, *cyt.* cytogenetics, *CLL* chronic lymphocytic leukemia, *NHL* Non-Hodgkin's lymphoma, *MDS* myelodisplastic syndrome, *HL* Hodgkin's lymphoma, *DLBCL* diffuse large B-cell lymphoma, *MCL* mantle cell lymphoma, *CR1* first complete remission, *CR* complete remission, *PR1* first partial remission, *RIC* reduced-intensity conditioning, *PR* partial response, *MAC* myeloablative conditioning, *AP* accelerated phase, *BP* blast phase

The EBMT Risk Score (Gratwohl et al. 1998, 2009)

This predictive score, validated with 56,505 patients, allows estimating the 5-year probability of OS and the treatment related mortality (TRM) for the main diseases (see Tables 11.2, 11.3, and 11.4).

The EBMT risk score is also useful for predicting OS and TRM in patients receiving a second HCT (Rezvani et al. 2012) and in those receiving a T-cell depletion (TCD) HCT (Lodewyck et al. 2011).

Some authors have introduced modifications in this risk score (including the concept of disease stage) to improve its predictivity (Terwey et al. 2010; Hemmati et al. 2011). Similarly, it has been combined with the HCT-CI (Barba et al. 2014).

Table 11.2 EBMT risk score (Gratwohl et al. 2009)

Variables	Value of variables	Points
Age	<20 Years	0
	20–40 Years	1
	>40 Years	2
Disease status[a]	Early	0
	Intermediate	1
	Advanced	2
Interval diagnosis of HCT[b]	<12 Months	0
	≥12 Months	1
Donor	HLA-identical sibling	0
	Unrelated donor	1
Gender of donor–recipient	Female to male	1
	Other combinations	0

Adapted from Gratwohl et al. (2009)
[a]Does not apply in patients with SAA. Early = AL in CR1; MDS in CR1 or untreated; CML in the first chronic phase; NHL/MM untreated or in CR1. Intermediate = AL in CR2; CML with another status than accelerated phase or blastic phase; MDS in CR2 or in PR; NHL/MM in CR2, PR, or stable disease. Late = AL in other stages; CML in blastic crisis; MDS in all other stages; NHL/MM in all other stages
[b]Does not apply to patients in CR1

Table 11.3 Probability (%) of TRM at 5 years by applying the EBMT risk score

Points	0	1	2	3	4	5	6–7
AML	14	20	25	30	36	40	41
ALL	15	23	24	30	40	47	53
CML	15	22	30	38	45	52	55
AA	18	26	40	49	52		
MDS	25	28	30	35	38	46	50
MM			29	35	40	42	52
NHL	15	24	28	30	34	36	38

Extracted from Gratwohl et al. (2009)

Table 11.4 Probability (%) of OS at 5 years by applying the EBMT risk score

Points	0	1	2	3	4	5	6–7
AML	68	59	52	38	30	23	18
ALL	66	52	43	38	22	16	14
CML	76	72	60	51	39	26	14
AA	81	72	60	49	45		
MDS	56	52	46	40	35	28	25
MM			48	40	36	22	17
NHL	75	59	50	48	43	40	38

Extracted from Gratwohl et al. (2009)

This score has been validated by many groups and for many diseases (AML, ALL, myelofibrosis (MF), CLL, and CML, among others).

HCT-Specific Comorbidity Index (HCT-CI) (Sorror et al. 2005)

This was developed in Seattle in 2005. It is an adaptation of the classical Charlson Comorbidity Index (CCI). It has been validated in several cohorts and is widely used. The score analyzes 17 comorbidities and their degree (see Table 11.5).

Table 11.5 HCT-specific Comorbidity Index, including age variable (Sorror et al. 2005, 2014)

Comorbidity/definition	Points
Age ≥ 40 years	1
Arrhythmia Atrial fibrillation, flutter, sick sinus node syndrome, or ventricular arrhythmias	1
Cardiac Coronary heart disease, congestive heart failure, IAM, FEVE ≤50%	1
Inflammatory bowel disease Crohn's disease or ulcerative colitis that has required treatment	1
Diabetes Requiring insulin or oral antidiabetic medication in the 4 weeks prior to HCT	1
Cerebrovascular CVA or TIA or cerebral thrombosis	1
Psychiatric Depression, anxiety, or others requiring ongoing treatment (not on demand)	1
Mild liver Chronic hepatitis, elevated bilirubin <1.5 × UNL or AST/ALT <2.5 × UNL Previous HBV or HCV infection	1
Obesity BMI >35 kg/m^2	1
Previous infection Infection in admission requiring continuation of treatment beyond day 0	1
Moderate lung DLCO and/or FEV1 66–80% or minimal stress dyspnea	2
Rheumatology Systemic lupus, rheumatoid arthritis, polymyositis, polymyalgia rheumatica, connective tissue disease	2
Peptic ulcer Endoscopic or radiological diagnosis (does not score if only reflux or gastritis)	2
Renal Creatinine >176 mcmol/L, dialysis, or previous kidney transplant	2
Previous tumor[a] Neoplasia at some point (excludes non-melanoma skin tumor)	3
Heart valve Diagnosed (except mitral prolapse)	3
Severe pulmonary DLCO and/or FEV1 ≤ %, dyspnea at rest or oxygen at home	3
Severe liver disease Bilirubin ≥0.5 for VN or AST or ALT ≥0.5 for VN or cirrhosis	3

[a]The most recent version has also included in this category hematological tumors of a different lineage, which motivates transplantation (e.g., lymphoma in an AML patient but not previous MDS in an AML patient)

CVT cerebral vein thrombosis, *TVA* transient ischemic attack, *AST* aspartate transaminase, *ALT* alanine transaminase, *BMI* body mass index, *FEVE* left ventricular ejection fraction, *IAM* myocardial infarction, *UNL* upper normal value

Given the impact of age on outcomes, the authors modified the model (Sorror et al. 2014), including a 1-point score for patients aged 40. This modification significantly improved the predictive capacity of the model. Consequently, the patients could be classified into three different risk groups (0 points: low risk; 1–2 points: intermediate risk; and 3 or more points: high risk) that clearly correlated with 2-year non-relapsed mortality (NRM).

Other authors re-stratified the HCT-CI index (flexible HCT-CI) as low risk: 0–3 points; intermediate risk: 4–5 points; and high risk: >5 points, with this classification being a better predictor of NRM. In an RIC setting, the 100-day and 2-year NRM incidence in these risk categories was 4%, 16%, and 29% and 19%, 33%, and 40%, respectively. The authors found this predictive NRM value using neither the original HCT-CI nor the pre-transplant assessment of mortality (PAM) or Charlson Comorbidity Index (CCI) models. Regarding the 2-year OS, this flexible HCT-CI score was also associated with the highest predictive hazard ratio (Barba et al. 2010).

HCT-CI has also been validated in CD34$^+$-selected HCT (Barba et al. 2017) and combined with the EBMT risk score, thus allowing a higher discrimination (Barba et al. 2014; Versluis et al. 2015).

The EBMT Machine Learning Algorithm (Shouval et al. 2015)

This algorithm is based on an alternating decision tree that is able to detect variables associated with the primary outcome, assigning weights and ignoring redundancies. This score was developed to not only analyze the NRM at day +100 in patients with acute leukemia but also predict NRM, Disease Free-Survival (DFS), and OS at 2 years.

The variables included in the model are age, Karnofsky index (≥80; <80), diagnosis (AML; ALL), disease stage (CR1; CR2; all other stages), interval diagnosis of HCT (<142 days; ≥142 days), donor–recipient CMV status (both sero+; both sero-; any other combination), donor type (matched sibling donor (MSD); matched unrelated donor (MUD)), conditioning (MAC; RIC), and annual allo-HCT performed at the center (<20; ≥21). The total + 100 NRM and 2-year NRM, DFS, and OS could be obtained through the web page http://bioinfo.lnx.biu.ac.il/~bondi/web1.html.

Recently, this algorithm has also been validated by an independent set of data from GITMO (Gruppo Italiano Trapianto di Midollo Osseo) (Shouval et al. 2017).

The Myelofibrosis Transplant Scoring System (MTSS) Score for Myelofibrosis

A clinical molecular myelofibrosis transplant scoring system (MTSS) has been proposed for patients for whom an allogeneic transplant is advised. This score is applicable to primary and post-essential thrombocythemia/polycythemia vera (ET/PV) myelofibrosis and seems promising in predicting posttransplant outcome better than disease-specific systems. Risk factors were incorporated into a four-level MTSS with a 5-year survival rate of 83% for low-risk patients (score, 0–2), 64% for intermediate-risk (score, 3–4), 37% for high-risk (score, 5), and 22% for very high-risk (score, >5) (Gagelmann et al. 2019).

11.1.4.3 The Predictive Capacity of these Models

Unfortunately, all these models have a limited predictive capacity, and none of them stand out more than the rest.

Authors	Predictive/s model/s	Predictive capacity
Sorror et al. (2005)	HCT-CI	0.65
Xhaard et al. (2008)	rHCT-CT, PAM	0.49, 0.57
Gratwohl et al. (2009)	EBMT	0.63
Barba et al. (2010)	fHCT-CI, PAM	0.67, 0.63
Barba et al. (2014)	HCT-CI, EBMT	0.60, 0.54
Versluis et al. (2015)	(HCT-CI-EBMT)	0.58, 0.58 (0.64)

rHCT-CI reduced model, without PFTs, HCT-CI flexible model (modified scale)

11.1.5 Practical Applications of Risk Assessment

Election of the conditioning	In patients with a high risk of NRM following one of the mentioned risk scores, an RIC should be considered
Relative contraindications	Uncontrolled infection, severe or chronic liver disease (excluding cirrhosis), severe disturbances in heart function (FEV <40%), respiratory (DLCO <40%), or renal (creatinine clearance <30 mL/min). *Cirrhosis:* Even compensated cirrhosis receiving RIC has a high likelihood of hepatic decompensation (Hogan et al. 2004)
Absolute contraindications	Pregnancy

Key Points

- Evaluation of a candidate must be carried out according to a preestablished work plan designed by each institution. The use of standardized procedures reduces the risk of errors or omissions.
- Pretransplant variables (such as age) have a clear impact on the results of the procedure but, when assessed in isolation, are highly insufficient to predict the results.
- Models (DRI, EBMT risk score, HCT-CI, PAM) allow a much more realistic approach to the real possibilities of a given candidate and adapt the procedure to their needs.

11.2 Counseling of Candidates

Alessandro Rambaldi

11.2.1 Introduction

Allo-HCT is a potentially curative treatment modality for otherwise incurable diseases. Unfortunately, after transplantation, patients may experience not only the persistence or recurrence of their own disease but also some dramatic clinical complications and toxicities, including death. The clinical indications for transplant have been addressed in the section "Indications" of this book, but, in general, when allo-HCT is advised, the strength of the indication (the likelihood to be cured by transplant), the patient's fitness, and his/her personal commitment to transplantation must be carefully evaluated for each candidate.

Obviously, in a patient, the first distinction must be made between neoplastic and non-neoplastic disease, and the transplant option should be progressively discussed with the patient during the course of the disease, particularly in the case of hematological malignancies. Many professionals should concur to illustrate to the patients the curative potential of allo-HCT and help them understand the severe complications that can eventually develop. It is clear that different indications remarkably affect the way in which a patient is advised. However, there is a time when the transplant option must be formally presented and advised. Therefore, evaluation of each transplant candidate must be based on well-predefined formal standard operating procedures to collect exhaustive clinical, instrumental, and laboratory data that may lead to a robust definition of the risks and benefits related to allo-HCT. All in all, counseling is tailored to conduct such evaluation of the individual patients (Shouval et al. 2015), according to both objective data and subjective data such as patient propensity and fear of side effects. At the end of this process, the patient should be aware of the rationale, the benefit and toxicity associated with each step, and the component of the transplantation procedure. In this chapter, I will hereby summarize the main topics that I cover with my patients when they come to my office to discuss the option of allo-HCT.

11.2.2 Understanding the Benefits and Risks of Allogeneic Transplantations

Patients must be informed that allo-HCT is a therapeutic option that is always proposed with the intent to achieve a permanent cure of the underlying disease, but, despite this premise, disease progression or relapse may eventually happen. The indication for allo-HCT depends not only on the disease characteristics but also on patient-related factors such as age and comorbidities (Sorror et al. 2007) so that the transplant proposal is the result of an accurate and wise evaluation of both these factors (Sorror 2013; Wang et al. 2014). In addition, the source of the stem cell graft and the donor type (sibling matched, matched or mismatched unrelated, haploidentical or cord blood) are associated with different transplant modalities for the graft procurement, the conditioning regimen, and the GvHD prophylaxis. All these details require time that the physicians must dedicate to their transplantation candidates.

The patient should understand the specific risk/benefit balance associated with a conventional versus a transplant-based proposal, and this may be remarkably different if he/she has been diagnosed with a non-neoplastic disease such as thalassemia or sickle cell anemia, a bone marrow failure syndrome like aplastic anemia, or a blood cancer, such as an acute leukemia. Even when allo-HCT may, in theory, represent the most efficacious treatment modality to get a permanent cure of a specific disease, an accurate description of the available alternatives must be presented. This is particularly important when the non-transplant options, albeit not curative, may have the chance to keep the patient alive for a long time (Samuelson Bannow et al. 2018) or, even more importantly, when conventional treatment may lead to a definitive cure such as in the case of some patients with acute leukemia with

intermediate-risk genetic factors or those achieving deep molecular remission after conventional chemotherapy (Cornelissen and Blaise 2016).

11.2.3 Understanding the Transplant Procedure: The Donor, the Conditioning Regimen, and the Clinical Complications

Once the indication for transplantation has been confirmed, patients and their relatives must be informed on how the transplant is performed. Patients should understand that identifying a stem cell donor is an absolute prerequisite to perform a transplant. Accordingly, patients should be informed about the human leukocyte antigen (HLA) genetic system, its specificity toward each individual, how it is inherited by parents according to the Mendelian laws, and what is the probability of finding a compatible donor in the family group. Understanding the HLA system is crucial to explain why the use of an HLA family-matched sibling donor is considered standard and when such a sibling is not available, an international search has to be performed to identify an HLA-compatible unrelated donor. It is important to underline that more than 40 million potentially available donors are registered by the World Marrow Donor Association (WMDA) and that the probability of finding a compatible donor is between 50 and 80% according to the ethnical origin of each patient.

Once such a matched unrelated donor is identified, this type of transplant is considered a standard of care, and its clinical outcome is fully comparable to that observed when using an HLA-identical sibling. In patients for whom an MSD or an MUD is not available, the patient should be informed that two additional options are available, namely, the use of HSCs obtained by a family mismatched donor (commonly defined as haploidentical because sharing only one of the patient's HLA haplotypes) or banked cord blood units (CBUs). Patients should understand how HLA diversity between the patient and donor has been overcome by specific programs of in vitro or in vivo manipulation of the graft. In recent years, the number of transplantations performed using an haploidentical donor has markedly increased since the clinical outcomes of this transplant modality are comparable to those achieved using matched related donors. Prospective clinical trials are ongoing to define if a MUD or an haplo donor can offer a clinical benefit, particularly in terms of leukemia-free survival. For this reason, dedicated counseling about the choice between an haplo and a MUD is frequently requested. Based on ethical considerations and the related probability of finding a donor within the international registries and also financial aspects, each transplant center must have standard operating procedures detailing the reasons why an haplo or a MUD donor is the preferred choice. The transplant candidate should be advised accordingly. In the current clinical practice, the use of cord blood units is declining and it is usually restricted to a minority of patients for whom a MUD or an haplo graft is not available. Still, the potent anti-leukemic activity of this transplant modality should be kept in mind, particularly for patients with resistant hematological malignancies (Horgan et al. 2023). Specific counseling to illustrate the benefits and risks (particularly the delayed engraftment and immune reconstitution) of this transplant modality should be undertaken.

Patients should be reassured that the incidence and severity of GvHD, the most important side effect of allo-HCT, does not seem to be higher than observed with a MUD and also an haplo donor. In addition, patients should know that many studies reported that transplants performed with these alternative stem cell sources proved to be effective and safe even when offered to patients of advanced age and/or with existing accompanying illnesses or when the disease was refractory to conventional treatment. All in all, at the present time, the clinical outcome of these alternative types of transplants compares reasonably well with those achieved with a MUD. Therefore, the decision to use this type of stem cell source only when an HLA-matched donor is not available is mostly related to the lack

of randomized clinical trials that are planned to be performed in the near future.

The goal of allo-HCT is to eradicate the patient's hematopoiesis that is either neoplastic or normal. This is achieved by delivery of the conditioning regimen and by the lifelong in vivo role played by the donor's immune system. Most often, high doses of chemotherapy and/or radiation are included in the preparations, although remarkable differences exist depending on the disease needing transplant and patient tolerance. The patient should understand that the intensity of the conditioning regimen may be particularly important in the case of hematological malignancies when the aim to remove most of the neoplastic cells present in the patient's body is the first goal. However, to avoid at least part of the treatment toxicity, the intensity of the preparative regimen can be down-modulated, leading to the definition of this preparative regimen as non-myeloablative or reduced intensity. The depletion of the patient bone marrow stem cells induces prolonged pancytopenia and the need for donor-derived healthy stem cells to grow and establish a new blood cell production system.

Allogeneic HSCs, collected from the donor's BM or PB or a frozen CBU, are infused through the central venous catheter into the bloodstream: HCT is not a surgical procedure, and it is highly similar to receiving a blood transfusion. The stem cells find their way into the bone marrow and begin reproducing and growing new, healthy blood cells. It is extremely important to explain how the donor immune system will develop progressively after transplantation and will either play a crucial beneficial role against residual neoplastic cells or restore the immune competence against infections, but it could mediate the most harmful GvHD effect against the patient.

After the transplant, supportive care is provided to prevent and treat infections, side effects of treatments, and complications. Prolonged anemia, thrombocytopenia, and leukopenia can be dangerous and even life-threatening. A low platelet count can be potentially associated with bleeding in the lungs, GI tract, and brain. Leukopenia, including either a defect of neutrophils and lymphocytes, leads to the development of frequent infections, the most common clinical complications after transplantation. Infections can include not only bacterial, most likely when the patient has severe bone marrow suppression, but also viral and fungal pathogens. Infections can require an extended hospital stay, prevent or delay engraftment, and/or cause permanent organ damage. On average, the time to hematological engraftment (recovery of the neutrophil and platelet function) is about 2–3 weeks, but a protective recovery of the immune system can take months and sometimes years. High doses of chemotherapy and radiation can cause remarkable toxicities that include but are not limited to severe mucositis (inflammation of the mouth and GI tract) that favors bacterial translocation with related infections and GvHD and multi-organ failure of mainly the lungs, heart, liver, and kidneys.

Particular attention should be paid to the risk of graft failure that can occur early or late after transplantation. Graft failure is more frequent in some diseases such as myelofibrosis or as a result of infection or when the stem cell content of the graft is insufficient to guarantee a durable engraftment. Graft rejection can also happen after a reduced intensity conditioning regimen (when the immune system of the host is not completely eradicated and can actively reject the donor stem cells).

Finally, and most importantly, patients must be aware of what GvHD is, when and how it may develop, and why it represents the most serious complication of HCT, being not only life-threatening but also the principal reason of a long-lasting poor quality of life. Transplantation candidates should be aware that GvHD is the negative counterpart of the deep interaction of the donor immune system within the patient's body that at the same time may lead to a definitive cure of an otherwise incurable disease. In other words, when transplant is advised, patients must realize that they are accepting the possible onset of a chronic, often invalidating, autoimmune disease. GvHD can appear at any time after transplant.

Patients must be reassured that specific and validated protocols to prevent GvHD have been implemented. These programs to prevent GvHD prophylaxis may differ substantially based on the patient's age, the conditioning regimen, the type of disease, and, most importantly, the stem cell graft and HLA compatibility. Patients must understand that despite GvHD prophylaxis, unfortunately, GvHD may eventually occur, and, in this case, specific treatment programs exist. It is important to teach the patient that GvHD can be distinguished into an acute form, which usually develops within the first 100 days after transplantation, and a chronic form, which occurs later in the transplant course. Patients who develop acute GvHD are also more likely to develop the chronic form of GvHD. Patients must understand the importance of their compliance to all the treatments given posttransplant to prevent GvHD and how this is instrumental for a successful transplant. GvHD occurs when the donor's immune system reacts against the recipient's tissue. At variance to what happens after a solid organ transplant where the patient's immune system is driven to reject only the transplanted organ, in GvHD, the donor immune system can react against many different organs of the recipient. This is because the new cells do not recognize the tissues and organs of the recipient's body as self. Over time, owing to the effect of immunosuppressive drugs, a progressive tolerance can develop. The most common sites for GvHD are the GI tract, liver, skin, and lungs.

11.2.4 Posttransplant Treatment Options

An important new development in transplantation is represented by the opportunity to deliver additional posttransplant antitumor treatments. The patients should be aware that transplantation may represent an immunotherapy platform, which does not impede the administration of additional antineoplastic treatments. The latter can give time to the donor immune system to mount an effective graft-versus-leukemia activity.

11.2.5 Alternative Treatment Options to Autologous and Allogeneic Transplantation

Recent development of innovative cellular therapies can represent an effective treatment alternative to both autologous and allogeneic transplant. Autologous chimeric antigen receptor T cells (CAR-T) cells as third- or even second-line treatment for patients with relapsed or refractory large B-cell lymphoma resulted in significantly longer overall survival than standard care (Westin et al. 2023). The results achieved in clinical trials have been confirmed in the real-world setting (Jacobson et al. 2020).

The benefit of this treatment option must be discussed with each potential patient, taking into consideration the most commonly reported side effects (cytokine release syndrome and neurotoxicity) as well the high costs of this treatment not always fully covered by national health systems or private insurances.

CAR-T cells have also been shown to be effective in relapsed and refractory ALL, with impressive results achieved in clinical trials conducted in pediatric and adult patients (Maude et al. 2018; Shah et al. 2021) and confirmed in the real-life setting (Pasquini et al. 2020). Despite the unprecedent rate of Complete Remission (CR) rate achieved in advanced ALL, many patients can still relapse. Therefore, a subsequent, allogeneic transplantation remains a treatment option to be discussed with the patient and his/her relatives, particularly in the pediatric setting. The high costs of a sequential CAR-T cell treatment and allogeneic transplant may represent a financial burden that is not always affordable.

11.2.6 Logistics

After discharge from the transplant ward, patients are followed up in the outpatient clinic two to three times per week until day +100. Patients should be helped to realize how complex the transplant procedure is and that the time spent in the hospital represents only the first part of the treatment pro-

gram. All allo-HCT patients should ideally stay within 1 h of the hospital until it is about 3 months from the day of the transplant. Patients and their families should also realize that the overall recovery time varies from person to person and that, in general, this process takes about 1 year to be considered satisfactory. Allogeneic transplantation is therefore a long-lasting form of immunotherapy, and the interaction between the donor immune system and the patient requires careful and prolonged medical assistance, quite often lifelong.

Key Points

Counseling of patients should be carefully performed to inform candidates that:

- The specific characteristics of both the disease and patient are equally important to advise transplant.
- Allo-HCT is performed to cure otherwise incurable diseases.
- Despite transplantation, the disease may persist or relapse may occur.
- Transplantation can severely compromise the quality of life of patients.
- Transplantation is a form of immunotherapy requiring long-term follow-up care.
- CAR-T cells are an alternative to autologous or allogeneic transplantation.
- Transplantation is an immunotherapy platform that does not impede additional posttransplant options.
- Logistics are important to ensure adequate care and assistance.

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